

Question:

Which of the following factors is associated with a decreased risk of Alzheimer's?

Answer Choices:

- A. Being African or Hispanic American
- B. A lower level of education
- C. Living in a city
- D. Being married
- E. Having a family history of Alzheimer's
- F. High intake of saturated fats
- G. Older age
- H. Eating fish
- I. Being a smoker
- J. Drinking alcohol regularly

Reasoning + Answer:

<tool_call> reasoning process here

Alzheimer's disease risk factors include age, genetics, lifestyle, and socioeconomic factors. Let's evaluate each option:

A. Being African or Hispanic American – Studies show that certain racial and ethnic groups, including African Americans and Hispanic Americans, may have higher rates of Alzheimer's due to social determinants and access to care, so this is linked to increased risk.

B. A lower level of education – Lower education is associated with increased risk due to reduced cognitive reserve.

C. Living in a city – Urban living is not consistently linked to lower risk; some studies suggest urban environments may have higher stress or pollution, potentially increasing risk. No strong evidence supports reduced risk.

D. Being married – Multiple studies suggest that marriage is associated with reduced risk of dementia and Alzheimer's. Social engagement and emotional support likely contribute to cognitive health, improving brain resilience.

E. Having a family history of Alzheimer's – Strongly linked to increased risk, especially with genetic factors.

F. High intake of saturated fats – Associated with increased inflammation and risk of cardiovascular disease, which is a risk factor for Alzheimer's.

G. Older age – The strongest risk factor; incidence rises with age.

H. Eating fish – Fish, especially fatty fish rich in omega-3 fatty acids, is associated with brain health and lower risk of neurodegenerative diseases. This is a plausible protective factor.

I. Being a smoker – Smoking increases risk due to inflammation and vascular damage.

J. Drinking alcohol regularly – Excess alcohol consumption increases risk, though moderate intake may have some protective effects; but regular heavy intake is harmful.

Among all options, both D (being married) and H (eating fish) appear protective. However, the best supported and most consistently observed factor from public health research is that social engagement, including being married, is linked to reduced dementia risk via psychosocial and biological pathways.

Therefore, the most strongly associated factor linked to decreased risk of Alzheimer's is:

D. Being married.

<answer>D</answer>

